

Q1 - 25 June - Shift 2

Let $A = \{x \in \mathbb{R} : |x + 1| < 2\}$ and

$B = \{x \in \mathbb{R} : |x - 1| \geq 2\}$. Then which one of the following statements is **NOT** true ?

(A) $A - B = (-1, 1)$ (B) $B - A = \mathbb{R} - (-3, 1)$

(C) $A \cap B = (-3, -1]$ (D) $A \cup B = \mathbb{R} - [1, 3)$

Space for your notes:

Q2 - 26 June - Shift 1

Let $A = \{n \in \mathbb{N} : \text{H.C.F.}(n, 45) = 1\}$ and

Let $B = \{2k : k \in \{1, 2, \dots, 100\}\}$. Then the sum of all the elements of $A \cap B$ is _____.

Space for your notes:

Q3 - 28 June - Shift 1

Let R_1 and R_2 be relations on the set $\{1, 2, \dots, 50\}$ such that

$R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$

and $R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$.

Then, the number of elements in $R_1 - R_2$ is _____.

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Q4 - 28 June - Shift 2

Let $R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}$. Then on $\mathbb{N}$:

(A) Both R_1 and R_2 are equivalence relations

(B) Neither R_1 nor R_2 is an equivalence relation

(C) R_1 is an equivalence relation but R_2 is not

(D) R_2 is an equivalence relation but R_1 is not

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Q5 - 29 June - Shift 1

Let a set $A = A_1 \cup A_2 \cup \dots \cup A_k$, where

$A_i \cap A_j = \phi$ for $i \neq j$, $1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y): y \in A_i \text{ if and only if } x \in A_i, 1 \leq i \leq k\}$. Then, R is :

- (A) reflexive, symmetric but not transitive
- (B) reflexive, transitive but not symmetric
- (C) reflexive but not symmetric and transitive
- (D) an equivalence relation

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Q6 - 29 June - Shift 2

The probability that a relation R from $\{x, y\}$ to $\{x, y\}$ is both symmetric and transitive, is equal to:

- (A) $\frac{5}{16}$
- (B) $\frac{9}{16}$
- (C) $\frac{11}{16}$
- (D) $\frac{13}{16}$

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Questions

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Answer Key

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Q1 (B)

Q2 (5264)

Q3 (8)

Q4 (B)

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Q5 (D)

Q6 (A)

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Q1 (B)

$$A: x \in (-3, 1) \quad B: x \in (-\infty, -1] \cup [3, \infty)$$

$$B - A = (-\infty, -3] \cup [3, \infty) = \mathbb{R} - (-3, 3)$$

Q2 (5264)Sum of elements in $A \cap B$

$$\begin{aligned}
 &= \underbrace{(2+4+6+\dots+200)}_{\text{Multiple of 2}} - \underbrace{(6+12+\dots+198)}_{\text{Multiple of 2 \& 3 i.e. 6}} \\
 &\quad - \underbrace{(10+20+\dots+200)}_{\text{Multiple of 5 \& 2 i.e. 10}} + \underbrace{(30+60+\dots+180)}_{\text{Multiple of 2, 5 \& 3 i.e. 30}} \\
 &= 5264
 \end{aligned}$$

Q3 (8)Here, $p, p^n \in \{1, 2, \dots, 50\}$ Now p can take values

2, 3, 5, 7, 11, 13, 17, 23, 29, 31, 37, 41, 43 and 47.

 $\therefore$ we can calculate no. of elements in R , as $(2, 2^0), (2, 2^1) \dots (2, 2^5)$ $(3, 3^0), \dots (3, 3^3)$ $(5, 5^0), \dots (5, 5^2)$ $(7, 7^0), \dots (7, 7^2)$ $(11, 11^0), \dots (11, 11^1)$

And rest for all other two elements each

$$\therefore n(R_1) = 6 + 4 + 3 + 3 + (2 \times 10) = 36$$

Similarly for R_2 $(2, 2^0), (2, 2^1)$ $(47, 47^0), (47, 47^1)$

$$\therefore n(R_2) = 2 \times 14 = 28$$

$$\therefore n(R_1) - n(R_2) = 36 - 28 = 8$$

Q4 (B)

$$R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$$

$$R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}.$$

For R_1 :

i) Reflexive relation
 $(a, a) \in \mathbb{N} \times \mathbb{N} : |a - a| \leq 13$

ii) Symmetric relation
 $(a, b) \in R_1, (b, a) \in R_1 : |b - a| \leq 13$

iii) Transitive relation
 $(a, b) \in R_1, (b, c) \in R_1, (a, c) \in R_1 :$

$$(1, 3) \in R_1, (3, 16) \in R_1, \text{ but } (1, 16) \notin R_1$$

For R_2 :

i) Reflexive relation
 $(a, a) \in \mathbb{N} \times \mathbb{N} : |a - a| \neq 13$

ii) Symmetric relation
 $(b, a) \in \mathbb{N} \times \mathbb{N} : |b - a| \neq 13$

iii) Transitive relation
 $(a, b) \in R_2, (b, c) \in R_2, (a, c) \in R_2$

$$(1, 3) \in R_2, (3, 14) \in R_2, \text{ but } (1, 14) \notin R_2$$

Q5 (D)

$$A = \{1, 2, 3\}$$

$$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Q6 (A)

$$\text{Total no. of relations} = 2^{2 \times 2} = 16$$

$$\text{Fav. relation} = \phi, \{(x, x)\}, \{(y, y)\}, \{(x, x)(y, y)\}$$

$$\{(x, x), (y, y), (x, y)(y, x)\}$$

$$\text{Prob.} = \frac{5}{16}$$